

Training Day 2 Report

Date: 18 June 2025

Module Name: Introduction to Information Security

Mode: Online

Main Focus of the Module

The second day of training focused on building a foundational understanding of **Information Security**. The session covered key theoretical concepts such as the **elements of information security**, types of **cyberattacks**, and the **vulnerabilities** that make systems prone to these attacks. We also explored how modern warfare has shifted to **information warfare**, and the distinction between **offensive** and **defensive cyber strategies**.

Topics Covered

1. Information Security Overview

- Ensures confidentiality, integrity, and availability (CIA Triad)
- Aims to protect data from unauthorized access, disclosure, disruption, or destruction

2. Elements of Information Security

- **Confidentiality** – Protecting sensitive data from unauthorized access
- **Integrity** – Ensuring data is accurate and unaltered
- **Availability** – Ensuring systems/data are accessible when needed
- **Authentication & Authorization** – Verifying identity and access rights
- **Non-repudiation** – Preventing denial of actions performed

3. Types of Attacks

- **Active Attacks** – Modify or destroy data (e.g., DoS, spoofing)
- **Passive Attacks** – Monitor data without altering it (e.g., sniffing)
- **Insider vs Outsider Attacks** – Origin of threat
- **Targeted vs Opportunistic Attacks** – Focused vs random attacks

4. Vulnerability and Its Causes

- **Definition:** A weakness in system design, configuration, or coding
- **Reasons for Vulnerability:**
 - Human errors
 - Unpatched software
 - Weak passwords
 - Misconfigured systems
 - Lack of security policies

5. Classification of Attacks

- **Network Attacks:** Sniffing, spoofing, DoS
- **Application Attacks:** SQL injection, XSS
- **Physical Attacks:** Theft of devices
- **Social Engineering:** Phishing, baiting, pretexting

6. Information Warfare

- Use of digital tools to gain advantage over adversaries
- Disruption of communications, misinformation, and system compromise

7. Defensive vs Offensive Cyber Warfare

- **Defensive Warfare:** Firewalls, IDS/IPS, patching, access control
- **Offensive Warfare:** Penetration testing, ethical hacking, red teaming
- Governments and organizations now invest in both strategies to prepare for cyber conflict

Key Takeaways

- Understood the core principles and pillars of information security
- Realized how even small vulnerabilities can lead to major data breaches
- Gained clarity on how attacks are classified and how modern warfare includes cyber dimensions
- Identified the growing need for both **proactive defense** and **controlled offensive security** in the cybersecurity landscape